LLARS: Enabling Domain Experts & Developers Collaboration for LLM Prompting, Generation, and Evaluation

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Abstract

We demonstrate LLARS, an open-source platform that bridges the gap between domain experts and developers for building LLM-based assistants. LLARS integrates three modules into an end-to-end pipeline: **Collaborative Prompt Engineering** for real-time co-authoring with version control and instant LLM testing, **Batch Generation** for systematic output production across prompts × models × data with cost control, and **Evaluation** for human and LLM assessors to rate, rank, label, or compare outputs with live inter-rater agreement metrics. Interviews with six domain experts and three developers in psychosocial counselling confirmed that LLARS eliminates the "translation work" between disciplines and enables domain experts to directly co-author prompts and conduct evaluations independently.

Demo: https://llars-project.github.io/demo

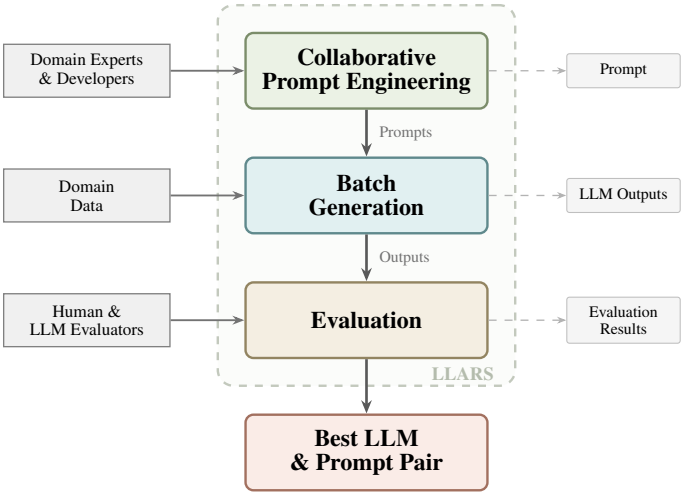

Figure 1: LLARS pipeline: domain experts and developers collaboratively develop prompts, generate outputs across LLMs, and evaluate results with human and LLM evaluators. Each stage supports export; the pipeline yields a validated model–prompt combination.

1 Introduction

Organizations building LLM-based assistants for sensitive domains [Chen *et al.*, 2024; Ling *et al.*, 2025]—psychosocial counseling [Stade *et al.*, 2024], legal analysis [Katz *et al.*, 2024], medical documentation [Van Veen *et al.*, 2024]—must write effective prompts, select suitable models, and rigorously evaluate outputs, requiring both domain expertise (understanding *what* constitutes quality) and technical skill (structuring prompts and configuring LLM pipelines) [Zamfirescu-Pereira *et al.*, 2023; Schulhoff *et al.*, 2024]. Yet these two groups typically work in isolation: prompts are iterated in shared documents without version control, and no single tool supports the full cycle from collaborative prompt development through systematic evaluation.

We present LLARS (LLM Assisted Research System), an open-source co-designed platform that closes this gap with three integrated modules (Figure 1). *Collaborative Prompt Engineering* lets domain experts and developers co-author prompts in a shared real-time editor with version control and instant LLM testing. *Batch Generation* systematically produces the Cartesian product of prompts × models × data items with cost estimation and budget control. *Evaluation* lets human and LLM evaluators assess outputs side by side—via rating, ranking, labeling, or pairwise comparison—with live agreement metrics (Krippendorff's α, Cohen's κ, and others). Each module can be used independently with JSON/CSV export, and the platform connects to any OpenAI-compatible provider including locally hosted models.

Table 1 surveys twelve existing platforms along these dimensions. Tracing-oriented tools (Phoenix [Arize AI, 2024], Langfuse [Langfuse GmbH, 2024], W&B Weave [Weights & Biases, 2024]) and prompt managers (Agenta [Agenta AI, 2024], ChainForge [Arawjo *et al.*, 2024]) provide per-output feedback but lack structured human evaluation with agreement metrics. Annotation platforms (Label Studio [HumanSignal, Inc., 2024], Argilla [Argilla, Inc., 2024]) offer structured assessment—Label Studio also computes basic agreement—but require externally generated outputs, disconnecting evaluation from prompt development. Closed-source tools (LangSmith [LangChain, Inc., 2024], Braintrust [Braintrust Data, Inc., 2024], Maxim [Maxim AI, 2024], Vellum [Vellum AI, 2024]) link prompting to evaluation but omit collaborative editing and batch generation as a pipeline

Table 1: Comparison of GUI-first tools for prompt engineering, batch generation, and hybrid LLM output evaluation. ● = native support; ◐ = partial; ○ = not supported. **Collab.** = real-time collaborative prompt co-editing; **Batch** = systematic generation over prompts × models × data; **H-Eval** = structured human evaluation campaigns; **LLM-J** = LLM-as-judge; **IRR** = inter-rater reliability metrics; **E2E** = integrated end-to-end pipeline.

Tool	OSS	Collab.	Batch	H-Eval	LLM-J	IRR	E2E
LLARS	●	●	●	●	●	●	●
Agenta [Agenta AI, 2024]	●	◐	◐	◐	●	○	◐
ChainForge [Arawjo *et al.*, 2024]	●	○	●	○	◐	○	◐
Phoenix [Arize AI, 2024]	●	○	◐	◐	●	○	○
Langfuse [Langfuse GmbH, 2024]	●	○	◐	◐	●	○	○
W&B Weave [Weights & Biases, 2024]	●	○	◐	◐	●	○	○
Label Studio [HumanSignal, Inc., 2024]	●	○	○	●	◐	◐	○
Argilla [Argilla, Inc., 2024]	●	○	○	●	◐	○	○
LangSmith [LangChain, Inc., 2024]	○	○	◐	◐	●	○	◐
Braintrust [Braintrust Data, Inc., 2024]	○	○	◐	◐	●	○	◐
Maxim [Maxim AI, 2024]	○	○	◐	◐	●	○	◐
Vellum [Vellum AI, 2024]	○	○	◐	◐	●	○	◐

stage. LLARS is the first open-source platform to unify collaborative prompt engineering, Cartesian-product batch generation, and hybrid human–LLM evaluation with agreement analytics.

2 Platform Overview

LLARS runs as a containerized web application connecting to any OpenAI-compatible LLM provider, including commercial services and locally hosted models. Teams register providers with their own API keys; models then appear throughout the platform.

2.1 Collaborative Prompt Engineering

The prompt engineering module (Figure 2) lets multiple users edit the same prompt simultaneously in a real-time collaborative editor built on Yjs [Jahns, 2023]. Prompts are structured as reorderable message blocks—system instructions, few-shot examples, user templates—with `{{variable}}` placeholders that the platform extracts into a shared palette for the team. Variables keep domain data separate from prompt logic: users assign sample data to each variable and can swap between inputs (e.g. two counselling threads) to verify generalisation instead of overfitting to a single example. When the prompt is ready, variables map directly to data columns in batch generation, so the same template processes an entire dataset without modification. Any collaborator can test the current prompt against a selected LLM while editing continues, and multiple team members can test different models or sample data in parallel. A version control system tracks every revision with multi-author attribution, supporting comparison and rollback. Finalized prompts are exported as JSON or passed directly to batch generation.

2.2 Batch Generation

The batch generation module (Figure 3) produces outputs at scale. A wizard walks users through selecting a data source (scenario, CSV/JSON upload, or none), choosing prompts and models, and setting budget limits. LLARS generates

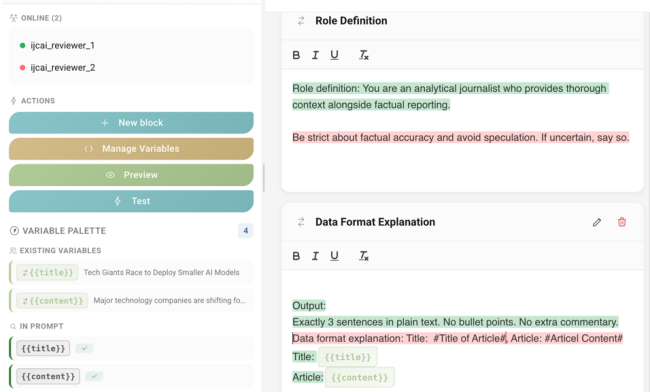

Figure 2: Collaborative prompt editor showing a news summarization prompt with two message blocks, template variables in the shared palette (left), and online presence. The sidebar provides testing, variable management, and JSON import/export.

the full Cartesian product—N data items × M prompts × K models—matching template variables to data columns automatically. Before execution begins, the platform estimates costs based on model pricing and token projections; optional budget caps pause jobs that exceed thresholds. Team members can monitor progress in real time as outputs stream in. Every result records its full provenance—source item, prompt, model, token count, and cost—and is exportable as CSV or JSON. Completed batches transfer directly into evaluation, preserving model attribution for cross-model comparison.

2.3 Evaluation

The evaluation module brings human and LLM evaluators together in a single framework. Four evaluation types are available (Figure 4): *rating* with configurable multi-dimensional Likert scales, *ranking* into ordered buckets, *labeling* for binary or multi-class classification, and pairwise *comparison*; the type system is extensible (Section 3). An AI-assisted

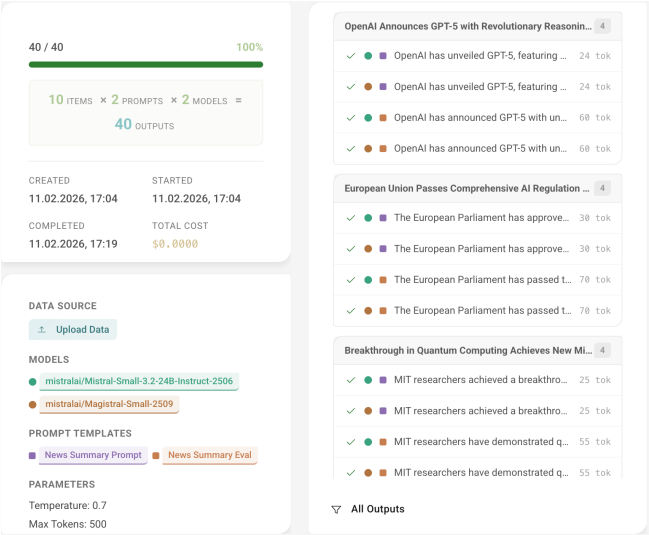

Figure 3: Batch generation results for a news summarization job: two prompts tested across two LLMs on ten articles, producing 40 outputs with full provenance. The *Scenario Wizard* button transfers outputs directly to evaluation.

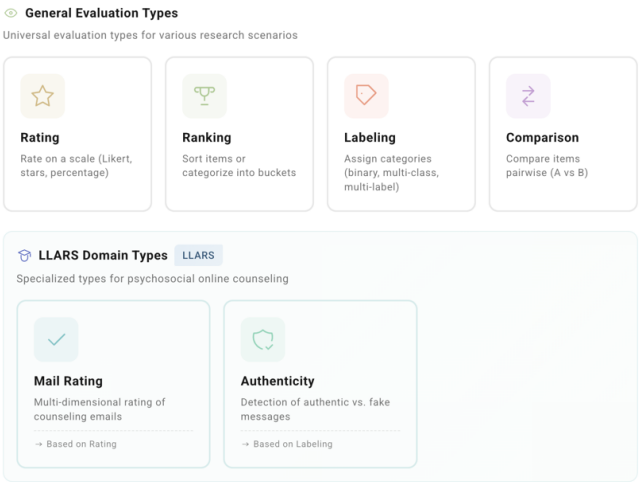

Figure 4: AI-assisted Scenario Wizard showing the task selection step with four general evaluation types and two domain-specific types for psychosocial counseling. The five-step wizard analyzes uploaded data and recommends the most suitable evaluation type.

Scenario Wizard analyzes uploaded data to recommend the most suitable type; scenarios can also be created directly from batch generation outputs, preserving model attribution.

Scenario owners control how items are distributed—all items to every evaluator for full inter-rater analysis, or configurable subsets for faster throughput. Where most tools limit LLMs to pairwise judging [Zheng *et al.*, 2024], LLARS treats them as full evaluation participants that rate, rank, label, and compare alongside humans using customizable type-specific prompts. Agreement metrics update live—Krippendorff's α, Cohen's and Fleiss' κ, Kendall's τ, and ICC—filterable by evaluator pairing (human–human, LLM–LLM, human–LLM). All data, metrics, and configurations are exportable as JSON or CSV.

3 Application and Preliminary Evaluation

LLARS is actively used in the development of CAIA [Steigerwald *et al.*, 2025], an AI-assisted system for text-based psychosocial counselling where LLM prompts transform unstructured conversations into structured outputs such as subject lines, case summaries, or response suggestions. Counselling researchers and developers collaboratively refined prompts for seven functionalities using LLARS, and the platform also motivated two domain-specific evaluation types (Figure 4): *Mail Rating* and *Authenticity*. For automated subject line generation [Steigerwald and Albrecht, 2025], the full pipeline was exercised end-to-end: batch generation produced 253 subject lines across 11 LLMs, and five professionals together with an LLM evaluator rated them in LLARS (1,518 assessments), directly informing model selection.

Semi-structured interviews with six domain experts (counselling researchers and practitioners) and three developers highlighted that the central benefit of LLARS is bridging the gap between disciplines. Domain experts reported that the collaborative editor eliminated the "translation work" previously required—formulating requirements in shared documents, exchanging them via e-mail, and having the development team convert them into technical formats. Instead, they could directly co-author and adjust prompts without needing to understand technical details, creating the experience of "truly working together on the same task." Developers valued the shared access point with version control. All domain experts rated the evaluation module as intuitive and a clear improvement over their previous Excel-based workflow; the drag-and-drop ranking was described as playful and more engaging than spreadsheet-based alternatives, and the progress tracking helped distribute evaluation work across multiple sessions.

4 Limitations, Demo, and Future Work

LLARS has so far been deployed in a single domain with a small user base, so broader studies are needed to assess generalisability. The quality of LLM evaluator judgments depends on careful prompt design, and the current evaluation relies on qualitative interviews rather than a controlled study. While the variable system helps prevent overfitting during prompt development, practitioners noted that prompts optimised on test cases can still underperform on unseen real-world data—a limitation inherent to any iterative prompt engineering process.

Our live demo walks attendees through the full pipeline, from collaborative prompt editing through batch generation to evaluation with live metrics. Future work targets a controlled user study comparing LLARS against baseline workflows (shared documents and spreadsheets) and cross-domain validation in legal and medical settings. LLARS is open-source and under active development; it continues to be deployed in ongoing projects to bring domain experts and developers together so that diverse fields beyond counselling can benefit from the current advances in AI.

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